

PRAFFUL KUMAR SINGH

Senior Software Engineer (FullStack)

Node.js | TypeScript | AWS | Serverless | Microservices | System Design

 Chandigarh, India (Open to Relocation) |  +91-9453667559

 praffulkumarsingh08@gmail.com |  linkedin.com/in/praffulks08051996

 github.com/praffulperplex

Professional Summary

- Senior Software Engineer with **5+ years of experience** designing and developing scalable, cloud-native backend applications using **Node.js, TypeScript, AWS, and Serverless Architecture**.
- Expertise in building **RESTful APIs, event-driven microservices, distributed systems, and secure healthcare platforms** compliant with **HIPAA and GDPR**.
- Hands-on experience with **AWS Lambda, API Gateway, DynamoDB, SQS, SNS, Step Functions, CloudWatch, IAM, EC2, and RDS**, delivering highly available, fault-tolerant, and cost-efficient solutions.
- Strong background in **system design, API architecture, database design, performance optimization, CI/CD automation, and third-party integrations** across healthcare, POS, and enterprise applications.
- Experienced in collaborating with cross-functional teams, mentoring engineers, driving technical decisions, and delivering high-quality software using Agile methodologies.
- Additional expertise in **Mobile Automation (Appium, WebdriverIO, TypeScript, Cucumber BDD)**, enabling end-to-end quality engineering across backend and mobile platforms.
- Additional expertise in **ReactJs** for developing **responsive user interfaces** and **Python** for backend development, **automation, scripting, and API integration**, complementing cloud-native application development.

Technical Skills

Backend Development & Cloud

- **Node.js + Python**
- JavaScript (ES6+) / TypeScript
- REST API Design & Development
- API Design Patterns
- Microservices Architecture
- Serverless Architecture (AWS Lambda, API Gateway)
- Event-Driven Architecture (SNS, SQS, Step Functions)

DevOps, Databases & Engineering

- Docker & Kubernetes
- CI/CD (GitHub Actions, Jenkins, AWS CodePipeline)
- Git & Version Control
- DynamoDB, MongoDB (NoSQL)
- MySQL (Relational Database)
- Data Modeling & Database Design
- Database Performance Optimization

Backend Development & Cloud

- Scalable System Design
- AWS Core Services
- Authentication (JWT, OAuth 2.0)
- Algorithm Optimization
- Strong Data Structures & Algorithms (DSA)
- **AI-Assisted Engineering (Cursor AI, ChatGPT, GitHub Copilot, ClaudeAI) – reducing development time by 30 to 40 %**
- Test Automation (Appium, WebdriverIO, BDD)
- **Python3 + React Experienced Developer**

DevOps, Databases & Engineering

- AWS EC2, RDS, CloudWatch, IAM, S3
- Third-Party Integrations (Payments, Printers, Invoicing, Maps)
- Agile/Scrum Methodologies
- Technical Mentorship
- Cross-Functional Collaboration
- Project Delivery & Time Management
- Data Analysis & Debugging
- Apache **Kafka** (proficient)

Professional Experience

Senior Software Engineer - [Aug 2024 to Present] | Firminiq Systems Pvt Ltd

- Designed and developed serverless backend services using AWS Lambda and API Gateway
- Built secure, scalable REST APIs **handling 10K+ requests/day, improving response time by ~30%** for mobile apps, dashboards, and partner integrations
- Implemented event-driven workflows reducing system latency by **~25%** and improving reliability using SNS, SQS, and Step Functions
- Designed and **optimized DynamoDB data models and access patterns**
- Implemented **OAuth 2.0 / JWT authentication** and role-based access control
- Secured APIs and data pipelines to meet **HIPAA** and **GDPR** requirements
- Enabled CloudWatch **logging, monitoring, and alerting** for production systems
- Automated deployments using **GitHub Actions, Jenkins, and AWS CodePipeline**
- Developed and maintained **React-based web applications integrated with backend APIs**

Backend Developer - [April 2023 to Jan 2024] | PoSist Technologies (New Delhi)

- Worked on **Complex data structures** and designed data models using **SQL** and **NoSQL** databases.
- **Experience in working on highly scalable cloud-based solutions, seamlessly integrating it with the existing POS system.**
- Utilized **SQL** for highly relational data and **NoSQL** for highly transactional data.
- Worked on **Hooks** and **APIs** to push or pull data within the system using **AWS Lambda**.
- Managed **API** responses and requests using Node Layer.
- Divided the whole system to **separate** data structures and **process-related functions, database-related functions (CRUD operations)** and **request/response handlers** within the system to reduce **Code redundancy**.
- Managed **documentation** for databases, APIs, and integrations.
- Leveraged **AWS** On-demand scaling to optimize server load and performance.
- Conducted comprehensive testing of **features, APIs, and SDKs**.
- Focused on maintaining **well-documented Code** quality, reducing **redundancy** following **DRY** principle.
- Leveraged **Git** for effective code version control and access management across the development team, ensuring code integrity and collaboration.

- Managed server deployments using **serverless architecture** for cost efficiency and rapid scaling.

Integrations / Backend - [April 2020 to March 2023] | Rocket Media Technologies Pvt Ltd

- Worked on **Third-party integrations**:
 - Payment providers**- CardConnect, Pinelabs
 - Delivery partners**- DoorDash
 - Printers** - Star Micronics Cloud Printer
 - Messaging Services** - WhatsApp, Twilio, AWS SNS, SQS.
 - Maps Integration** - Google, Mapbox
 - Helper Libraries** - Url Analytics, Order Receipt Integrations
- Assisted in monitoring **Data integrations, APIs, databases, and API documentation.**
- **Collaborated with Client Service Providers** in case of any issue or misconceptions faced while **Integrating Services** to get them resolved, if any.
- Experienced on **Working on highly complex menu structures** to adapt to diverse and complex menu configurations.
- **Collaborated directly with store managers** to understand challenges and enhance the functionality of existing software.
- **Improved user experience by optimizing workflows, reducing clicks, and enhancing software usability.**
- **Constructed SDKs for seamless integration into UI, fostering a cohesive user experience.**
- **Collaborated closely with the UI team, providing insights into data structures and requirements for effective system development.**
- Backend Server handling, **Logics Improvements** and **Code Optimizations.**

Projects

Remote Patient Monitoring (RPM) Platform – VitalSight

Enterprise-grade RPM platform enabling physicians to remotely monitor hypertension patients via **EMR integrations** and web dashboards.

Key Contributions:

- Developed scalable **Node.js microservices** to support real-time healthcare data processing across multiple regions.
- Integrated **Electronic Medical Record (EMR)** systems with secure REST APIs for seamless clinician data exchange.
- Built automated **alerting and notification pipelines** for abnormal patient vitals, improving timely clinical intervention.
- Ensured **data integrity, encryption, audit logging, and compliance** with healthcare security and regulatory standards (HIPAA/GDPR).
- Implemented and supported **patient enrollment, billing, and notification workflows** for remote patient monitoring services.
- **Collaborated with Product, QA, Compliance, and cross-functional teams** in an Agile/Scrum environment to deliver high-quality features.

KIOSK: (for Middle East and Indian Clients)

- Contributed to designing and developing the **backend architecture** for a cloud-based KIOSK solution from scratch, focusing on scalability, maintainability, and long-term extensibility.
- Designed efficient **data structures and backend business logic** using JavaScript, applying data structures and algorithmic approaches to optimize application performance.
- Developed reusable **JavaScript helper libraries** to simplify integrations, improve code reusability, and reduce development effort across the KIOSK platform.
- Built and integrated a **highly scalable cloud-based backend** with the existing POS ecosystem, ensuring seamless data synchronization and system interoperability.
- Engineered solutions to efficiently process and manage **complex menu structures**, improving performance and maintainability for diverse client configurations.
- Collaborated closely with **UI/UX and frontend teams** to design, integrate, and optimize REST APIs, ensuring reliable data exchange and a seamless user experience.

User Web Dashboard and Ecommerce Website: (for US Clients)

- Worked on **third-party integrations** including Payment Providers, Delivery Partners, Printers, Messaging Services, Maps Integration, and Helper Libraries.
- Assisted in monitoring **data integrations, APIs, databases, and technical documentation**.
- Collaborated directly with **store managers** to understand business challenges and enhance the functionality of existing software.
- Constructed **SDKs** for seamless integration with the UI, fostering a cohesive user experience.
- Collaborated closely with the **UI team**, providing insights into data structures and system requirements for effective application development.
- Implemented **backend server logic** and performed **code optimizations** to improve application performance and maintainability.

Achievements

- **Rising Star Award** - Recognized for outstanding technical contributions and project delivery (Firminiq)
- **Qualified IIT JEE Main examination in 2014 and 2015.**
- **Improved backend reliability through monitoring and automated testing**

Education

Bachelor of Technology - CSE (*Computer Science and Engineering*) **2015 - 2019**
Feroze Gandhi Institute of Engineering and Technology (Government College - AKTU)

Hobbies

- Playing Cricket and Team Sports
- Reading Technology & Personal Development Books
- Travelling and Exploring New Places
- Networking, Meeting New People, and Learning from Diverse Experiences